# Customer Identity Resolution for Multi-Source Customer Data

Design and operations based on the current schema and source code

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## Abstract

Customer Identity Resolution (CIR) is the process of linking customer records from multiple systems into a single unified profile. In this implementation, inbound data is first written to a staging table and then processed by a Python resolver driven by metadata rules. The objective is to maintain exactly one master profile per customer within each tenant and domain, while preserving flexibility when new data sources or matching rules are introduced.

## 1. Scope

This paper is based on components that already exist in the repository:

- PostgreSQL schema in `database-init/database-schema.sql`
- Resolver module in `backend-system/identity_resolution/identity_resolution/resolver.py`
- Trigger controller in `backend-system/identity_resolution/identity_resolution/trigger_controller.py`
- Persona logic in `backend-system/identity_resolution/identity_resolution/persona.py`

### Main Flow (Identity Resolution Context)

```
Raw Profiles
  -> Staging and Validation
  -> Identity Resolution Engine (this paper)
       -> Rule Loading from Metadata
       -> Candidate Matching (exact/fuzzy)
       -> Master Profile Merge or Create
       -> Link and Audit Updates
  -> Unified Customer Profile
  -> Persona Resolution Engine (downstream)
  -> Customer360 Master Profile
```

| Stage | Purpose | Output |
|---|---|---|
| Raw Profiles | Ingest events and profile traces from source systems | Staging-ready records |
| Identity Resolution | Match, merge, and link records per tenant/domain | Unified customer profile |
| Downstream Persona Resolution | Enrich profile with persona intelligence | Persona fields, scores, and embeddings |
| Customer360 Master Profile | Persist durable customer truth for activation | Single customer view for operations |

## 2. Core Data Model and Data Enrichment Structure

The database stores entities and relationships required by the resolution lifecycle. This section summarizes the core tables, key fields, and their operational role in enrichment.

### 2.1 Table `cdp_raw_profiles_stage` (Raw staging table)

This intermediate table acts as the landing zone for raw customer traces from heterogeneous source systems before identity resolution.

**Important fields:**

- `raw_profile_id` (UUID, primary key): unique identifier for each raw profile event.
- `tenant_id` (UUID): tenant partitioning key for multi-tenant isolation.
- `domain` (TEXT): business domain context (for example: `retail`, `banking`, `travel`).
- `source_system` (TEXT): upstream origin system (for example: `POS`, `AppsFlyer`, `MoEngage`, `CRM`).
- `external_customer_id` (TEXT): source-local customer identifier.
- `email`, `phone_number`, `national_id` (TEXT): personal identifiers; values can be plaintext or one-way SHA-256 hashes.
- `full_name`, `first_name`, `last_name` (TEXT): source name fields.
- `device_id`, `advertising_id`, `cookie_id`, `push_token` (TEXT): digital/device identifiers for cross-channel resolution.
- `event_name`, `event_time` (TIMESTAMPTZ), `event_payload` (JSONB): event semantics, timestamp, and extensible attributes for enrichment.
- `status_code` (SMALLINT, default `1`): queue lifecycle (`1` ready, `2` processing, `3` processed, `4` failed).
- `processed_at` (TIMESTAMPTZ): completion timestamp.

### 2.2 Table `cdp_master_profiles` (Master/Golden profile)

This table stores the canonical customer profile after merge, normalization, and enrichment across all sources.

**Important fields:**

- `master_profile_id` (UUID, primary key): golden profile identifier.
- `tenant_id` (UUID), `domain` (TEXT): strict data scoping dimensions.
- `full_name`, `first_name`, `last_name`, `email`, `phone_number`, `national_id`, `address` (TEXT): consolidated identity fields.
- `secondary_emails`, `secondary_phones` (JSONB): retained alternate contact points to preserve channel reach.
- `external_ids` (JSONB): source-to-external ID map to support reverse synchronization.
- `device_ids`, `advertising_ids`, `cookie_ids` (TEXT[]): deduplicated device identity arrays.
- `push_tokens` (JSONB): push-token map for notification platforms.
- `is_hashed` (BOOLEAN, default `FALSE`): indicates that sensitive PII values are hashed.
- `persona_name` (TEXT): non-PII identity label generated by deterministic logic or LLM.
- `persona_summary` (TEXT): short behavior narrative used in personalization workflows.
- `persona_embedding` (VECTOR(768)): embedding vector for semantic retrieval and lookalike targeting.
- `updated_at` (TIMESTAMPTZ): last update timestamp.
- `first_seen_raw_profile_id` (UUID): lineage reference to the first raw record that created this master profile.
- `source_systems` (TEXT[]): set of contributing source systems.

### 2.3 Table `cdp_profile_links` (Profile-link table)

This table stores the active 1-to-N relationship between a master profile and its contributing raw profiles.

**Important fields:**

- `link_id` (BIGSERIAL, primary key): unique link record identifier.
- `tenant_id` (UUID): tenant partition key.
- `raw_profile_id` (UUID, UNIQUE): one raw profile maps to one active master profile at a time.
- `master_profile_id` (UUID): target master profile.
- `match_score` (NUMERIC): confidence score in range [0.0, 1.0].
- `match_method` (TEXT): algorithm/method label (for example: `exact_email`, `fuzzy_trgm_name`).
- `status` (VARCHAR): link state (`ACTIVE`, `HISTORICAL`).

### 2.4 Table `cdp_profile_attributes` (Attribute metadata registry)

This registry controls matching and consolidation behavior at the attribute level.

**Important fields:**

- `attribute_internal_code` (VARCHAR, primary key): internal key aligned with staging columns.
- `master_profile_column` (VARCHAR): target column on `cdp_master_profiles`.
- `is_identity_resolution` (BOOLEAN): whether the attribute participates in identity matching.
- `matching_rule` (VARCHAR): matching method (`exact`, `fuzzy_trgm`, `fuzzy_dmetaphone`, `none`).
- `matching_threshold` (NUMERIC): threshold for fuzzy matching.
- `consolidation_rule` (VARCHAR): merge strategy (`most_recent`, `verified_first`, `source_priority`, `non_null`, `append_distinct`, `overwrite`).

### 2.5 Table `cdp_identity_index` (Flattened identity index)

This index accelerates exact-match lookups by flattening JSON/array identifiers, reducing scans over complex nested fields under high throughput.

**Important fields:**

- `identity_index_id` (UUID, primary key): identity index record identifier.
- `tenant_id` (UUID): tenant scope.
- `master_profile_id` (UUID): owner master profile.
- `identifier_type` (VARCHAR): identifier type (for example: `email`, `phone`, `cookie_id`).
- `identifier_value` (TEXT), `identifier_value_normalized` (TEXT): raw and normalized forms for stable exact matching.
- `is_blocked` (BOOLEAN): blocks invalid/synthetic values (for example: `anonymous`, `null`, `void`).

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## 3. Processing Mechanism

Customer Identity Resolution is orchestrated by `CustomerIdentityResolver` in `resolver.py`. The pipeline is metadata-driven through `cdp_profile_attributes`, which provides strong flexibility and maintainability while preserving strict multi-tenant isolation.

### 3.1 Detailed Step-by-Step Processing

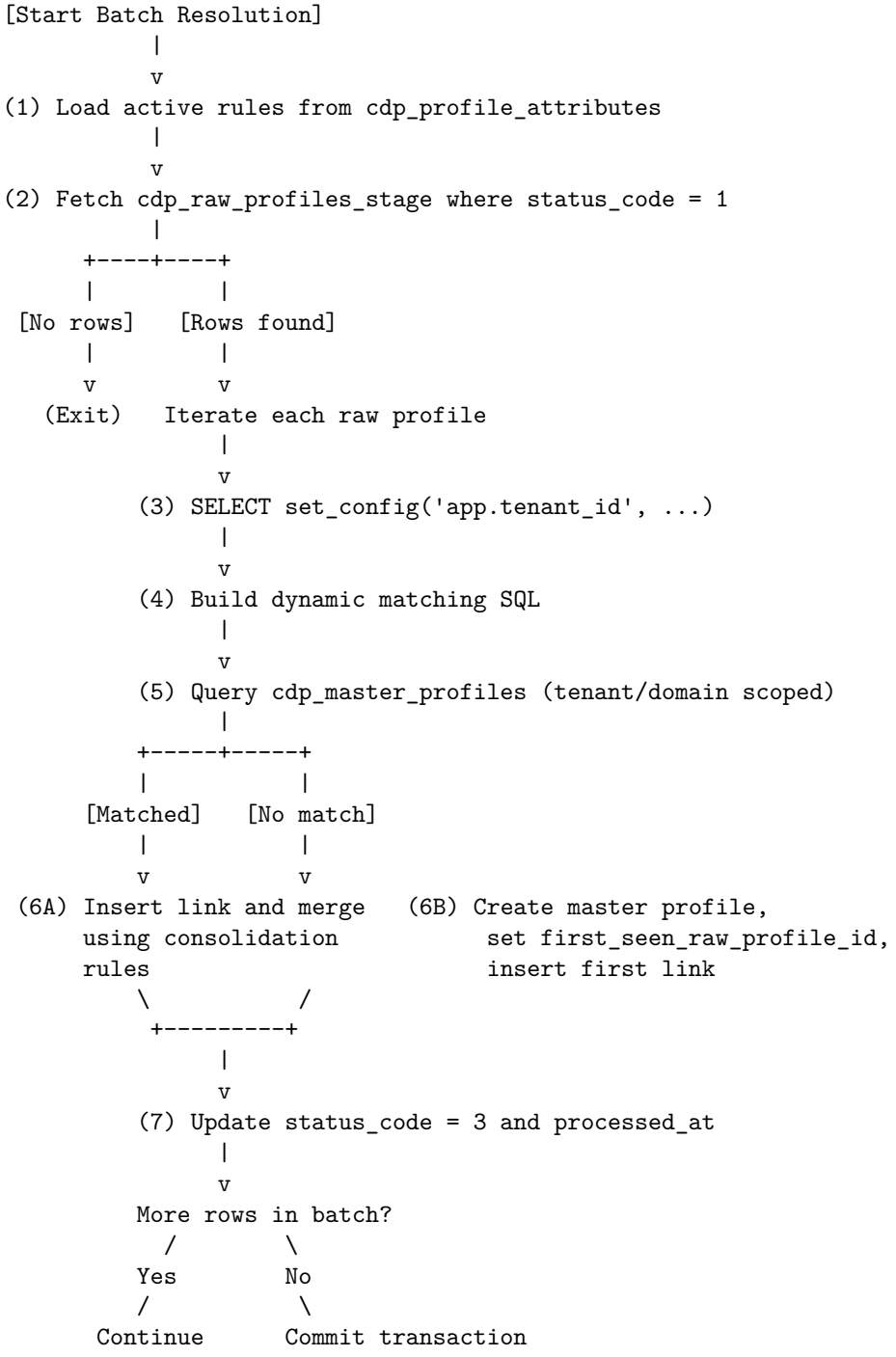
Each batch executes a sequential seven-step flow:

1. **Load active rules**
   - Query `cdp_profile_attributes` for attributes where `is_identity_resolution = TRUE`, `status = 'ACTIVE'`, and `matching_rule` is defined and not `none`.
2. **Fetch unprocessed staging records**
   - Query `cdp_raw_profiles_stage` with `status_code = 1`, limited by configured `batch_size` (default: 1,000 records).
3. **Set secure tenant context**
   - For each raw row, execute `SELECT set_config('app.tenant_id', ...)` to activate PostgreSQL RLS and prevent cross-tenant leakage.
4. **Build dynamic matching query**
   - Construct SQL `OR` conditions from available staged attributes:
     - Device arrays (`device_id`, `advertising_id`, `cookie_id`): `= ANY(array_column)`
     - Source-keyed IDs (`external_customer_id`): `external_ids @> jsonb_build_object(source_system, value)`
     - Exact match: `=`
     - Trigram fuzzy match: `similarity(column, value) >= threshold`
     - Phonetic fuzzy match: `dmetaphone(column) = dmetaphone(value)`
5. **Find candidate master profile**
   - Execute scoped SQL with mandatory partition filters:
     - `WHERE tenant_id = :tenant_id AND domain = :domain AND (dynamic_conditions) LIMIT 1`
6. **Merge or create master profile**
   - **If a master profile is found:**
     - Insert link in `cdp_profile_links` with `match_score = 1.0`, `match_method = 'DynamicMatch'`
     - Merge scalar attributes according to configured `consolidation_rule`
     - Union/append arrays and JSON maps (`source_systems`, `external_ids`, `push_tokens`, and related fields)
     - Detect hashed PII and, when needed, set `is_hashed = TRUE` and generate `persona_name`
   - **If no master profile is found:**

- Create new row in `cdp_master_profiles`, preserving `first_seen_raw_profile_id`
- Insert first link with `match_method = 'NewMaster'`

7. **Mark processed and commit**
   - Update staging row to `status_code = 3` and set `processed_at = NOW()`
   - Commit transaction at batch end; on failure, rollback

### 3.2 Resolution Execution Flow




### 3.3 Matching Rules and Data Examples

The system supports four rule families controlled by `matching_rule` in `cdp_profile_attributes`.

**1.** `exact` **rule** Used for stable high-precision identifiers. Matching uses `=` or array containment semantics where applicable.

- **Inbound staging:**
  - `email`: `nguyena@gmail.com`
  - `phone_number`: `+84901234567`

- **Existing master:**
  - email: nguyena@gmail.com
  - phone_number: +84901234567
- **Outcome:** deterministic match; row is linked to existing master profile.

**2. `fuzzy_trgm` rule** Used for typo-tolerant text fields such as names or addresses. It relies on PostgreSQL `pg_trgm` similarity with configurable threshold.

- **Inbound staging:**
  - full_name: Nguyen Van A
  - address: 123 Duong Le Loi, Phuong 1, Quan 1, TPHCM
- **Existing master:**
  - full_name: Nguyen Van A
  - address: 123 Le Loi, P.1, Q.1, TP HCM
- **Outcome:** similarity exceeds threshold (for example, 0.65), so records are treated as same customer.

**3. `fuzzy_dmetaphone` rule** Used for phonetic matching of names with small spelling variations.

- **Inbound staging:** first_name = Smith
- **Existing master:** first_name = Smyth
- **Outcome:** both map to equivalent phonetic code, producing a successful match.

**4. `none` rule** Attribute is excluded from identity matching and used only for post-match enrichment.

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## 4. Sensitive Data Handling (Privacy and Hashed PII)

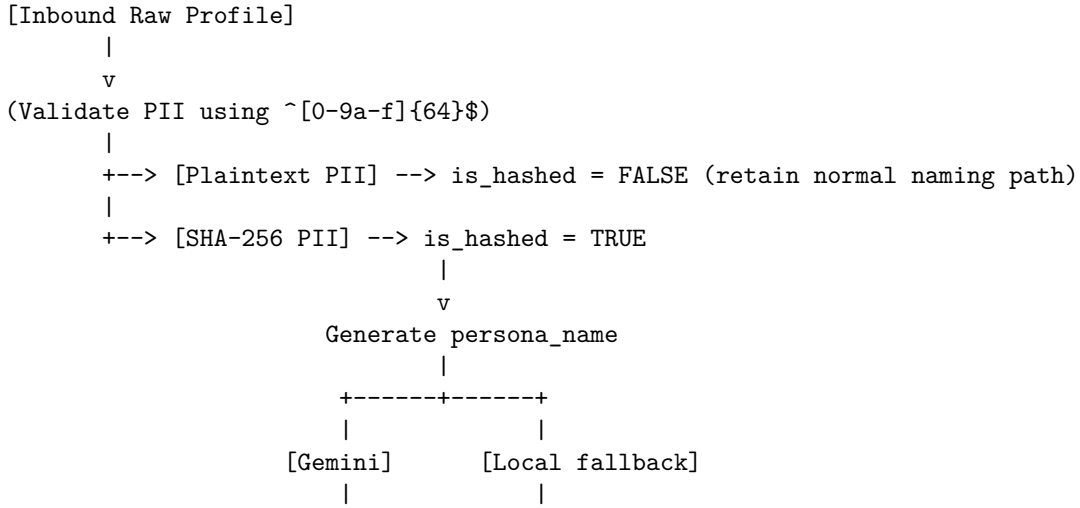
To align with personal-data regulations and ad-tech hashed matching patterns, the engine supports one-way SHA-256 PII payloads (64 hex characters).

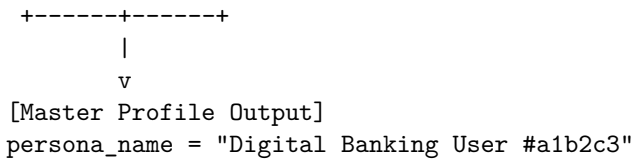
### 4.1 Detection and Persona Name Generation

When PII fields (`full_name`, `email`, `phone_number`, `national_id`) arrive as hashes, plaintext display is impossible. Therefore, `persona.py` triggers a non-PII naming path:

1. **Automatic detection**
   - Regex `^[0-9a-f]{64}$` identifies hashed values
   - `is_hashed = TRUE` is set on `cdp_master_profiles`
2. **Persona generation**
   - **LLM path (Gemini):** if `GOOGLE_GENAI_API_KEY` is configured, generate memorable non-PII names from non-sensitive context (domain, channel, source)
   - **Deterministic offline fallback:** if no API key/network, derive a stable label from anchor identifiers (`device_id`, `advertising_id`, and similar), with a 6-character hash suffix

### 4.2 Sensitive Data Sequence Flow






### 4.3 Example Data (Hashed vs Plaintext)

| Attribute | Plaintext Input | Hashed Input (SHA-256) | Stored in Master |
|---|---|---|---|
| full_name | Nguyen Van An | 9f86d08188...15b0f00a08 | 9f86d08188...15b0f00a08 |
| email | an.nguyen@gmail.com | d081884c7d...c15b0f00a08 | d081884c7d...c15b0f00a08 |
| is_hashed | FALSE | TRUE | TRUE |
| persona_name | NULL | auto-generated | Savvy Retail Shopper #4f2a9c |

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## 5. Triggering and Throttling

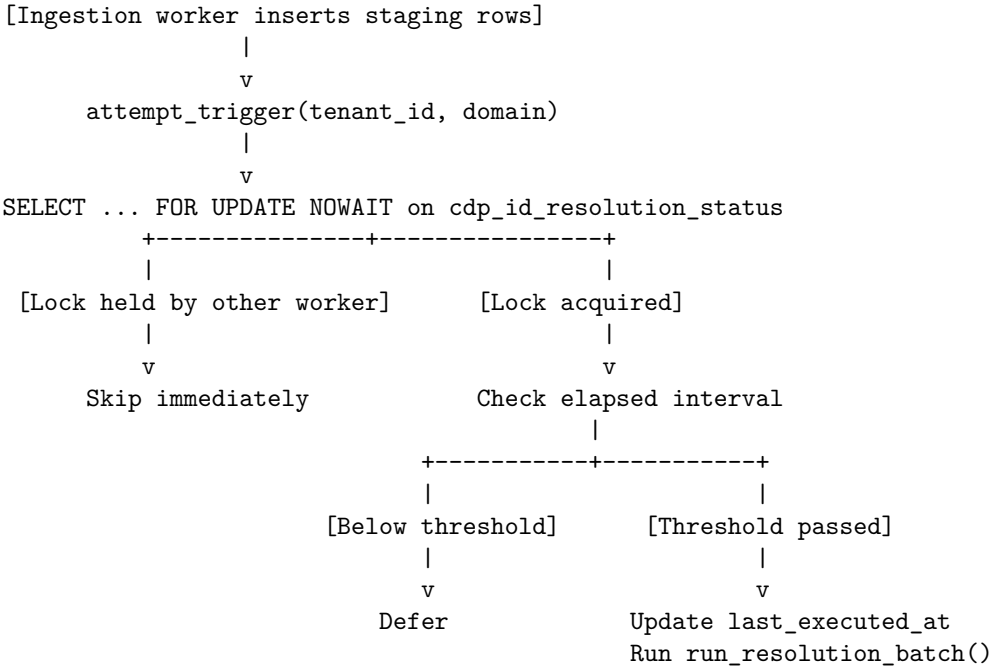
To sustain performance under continuous ingestion, the architecture combines near-real-time triggering with periodic batch processing.

### 5.1 Row-Lock Throttling Mechanism

The system avoids direct PL/pgSQL triggers to reduce lock contention. Instead, the ingestion worker invokes `IdentityResolutionTrigger.attempt_trigger()` after writes:

1. Runtime state is controlled by single-row table `cdp_id_resolution_status`.
2. `SELECT ... FOR UPDATE NOWAIT` is used:
   - If lock is held by another worker, current trigger call exits immediately (no blocking).
   - If elapsed time from prior run is below `throttle_seconds` (default: 5 seconds), trigger is deferred for micro-batching.
3. If conditions pass, worker updates `last_executed_at` and runs `run_resolution_batch()`.

### 5.2 Trigger and Throttling Control Flow



### 5.3 Operational Status Example

Table `cdp_id_resolution_status`:

| id | last_executed_at | updated_at | Runtime Interpretation |
|---|---|---|---|
| TRUE | 2026-08-04 10:00:00+07 | 2026-08-04 10:00:00+07 | Run at `10:00:02` is throttled (`< 5s`), run at `10:00:06` is accepted |

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## 6. Operational Best Practices

1. **Idempotency and integrity**
   - Process only staging rows with `status_code = 1`
   - Move processed rows to `status_code = 3` with `processed_at = NOW()`
   - `UNIQUE(tenant_id, raw_profile_id)` on `cdp_profile_links` prevents duplicate links on retries
2. **Tenant and domain security boundaries**
   - All generated SQL enforces `WHERE tenant_id = :tenant_id AND domain = :domain`
   - This prevents cross-tenant and cross-domain leakage
3. **Data normalization before matching**
   - Phone values should follow E.164 format (for example, `+84901234567`)
   - Email values should be lowercased and trimmed before staging
4. **Failure handling and queue monitoring**
   - Alert on excessive `status_code = 1` backlog
   - Alert on recurrent `status_code = 4` failures

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## 7. Conclusion and Solution Evaluation

The Customer Identity Resolution implementation in this Customer 360 architecture is metadata-driven, tenant-aware, and operationally practical. It supports both near-real-time micro-batching and scheduled high-throughput batch execution, while producing stable golden profiles without introducing database bottlenecks.

### 7.1 Use Case Application Matrix

| Business Use Case | Core Technical Requirement | C360 CIR Resolution Mechanism |
|---|---|---|
| Cross-channel identity stitching | Link Web/App/POS/CRM/Ads records | Exact/fuzzy matching on contact identifiers plus accumulation of `device_ids`, `cookie_ids`, `advertising_ids`, and `external_ids` |
| Multi-tenant and multi-domain governance | Strict data partition by tenant/domain | Hard SQL scope filters and PostgreSQL RLS |
| Privacy-preserving hashed PII and ad-tech matching | Support SHA-256 payloads without plaintext exposure | Hash detection, `is_hashed` flagging, non-PII persona naming via Gemini/local fallback, optional embedding enrichment |
| Dirty data and typo tolerance | Resolve identities under spelling variations | `fuzzy_trgm` and `fuzzy_dmetaphone` rules with configurable thresholds |
| Attribute conflict handling | Resolve contradictory source values | Field-level `consolidation_rule`: `verified_first`, `most_recent`, `source_priority`, `append_distinct`, and others |

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### 7.2 Detailed Comparison with Commercial Alternative (Twilio Segment Unify vs Native C360 CIR)

| Comparison Dimension | Native Customer 360 CIR Engine | Twilio Segment Unify (Personas) |
|---|---|---|
| Deployment and governance | Native PostgreSQL 16 (on-prem/private cloud), full infrastructure and data control, no vendor lock-in | Managed SaaS, data traverses vendor infrastructure and governance model |
| Matching engine | Hybrid matching: exact + trigram fuzzy + phonetic fuzzy, metadata-driven | Primarily deterministic identity graph around `userId`/`anonymousId`; limited native fuzzy matching |
| Multi-tenant isolation | Database-level isolation with RLS and strict tenant/domain scoping | Workspace/source-level separation; can become costly at large tenant counts |
| Consolidation strategy | Per-field configurable merge policies (`most_recent`, `verified_first`, `source_priority`, `append_distinct`, `overwrite`) | Simpler trait-priority/last-write defaults |
| Privacy and AI enrichment | Native SHA-256 handling, persona name generation with Gemini, embedding generation for semantic use cases | Usually requires extra custom transformations/functions outside core flow |
| Total cost of ownership | Primarily infra and worker cost; no MTU licensing growth curve | MTU-based pricing; cost scales rapidly with user volume |
| Ingestion latency and throughput | Flexible near-real-time micro-batching or scheduled batch | Strong event-driven SaaS near-real-time model |

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### 7.3 Final Summary

By combining metadata-driven flexibility, secure hashed-PII handling, and strict tenant isolation at the database layer, the Native C360 CIR Engine provides a strong and cost-efficient foundation for large-scale identity unification. The design is particularly suitable for organizations that require full data ownership, transparent matching logic, and low long-term TCO while still supporting advanced personalization and analytics workflows.